Panskura Banamali College(Autonomous)

B.Sc.(H)6th Semester Examination -2022 (Online Mode)

Subject: Computer Science **Paper:DSE3T**

PaperName: Machine Learning

F.M.:40 **Time:2Hours**

Answer any four : **4x10=40**

1. Describe Backpropagation algorithm with proper example. 10
2. Explain SVM algorithm in detail with example. 10
3. a)How Supervised Learning is different from Unsupervised Learning techniques?
   b)How regression is different from classification?
   c)Define Information gain. 5+3+2
4. a)What is Overfitting?How Can You Avoid It?
   b)Write down differences between training set and test set. 5+5
5. a)How Can we represent a supervised learning model?
   b)How does a Naive Bayes classifier help in classifying the output class?
   c)Define Bays theorem. 3+7
6. a)Explain the differences between classification and regression.
   b)What is clustering?
   c)How can we select K for K-means clustering algorithm? 5+2+3

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